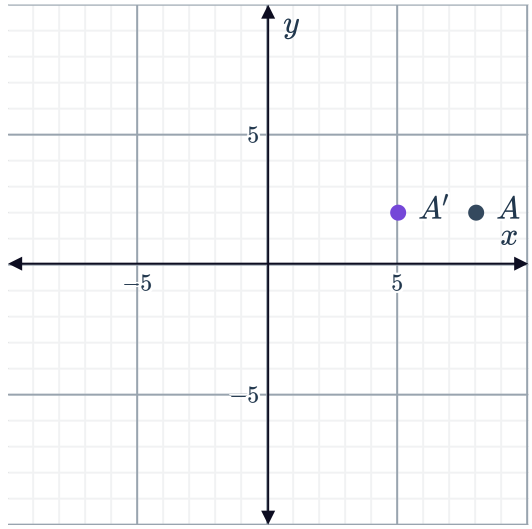
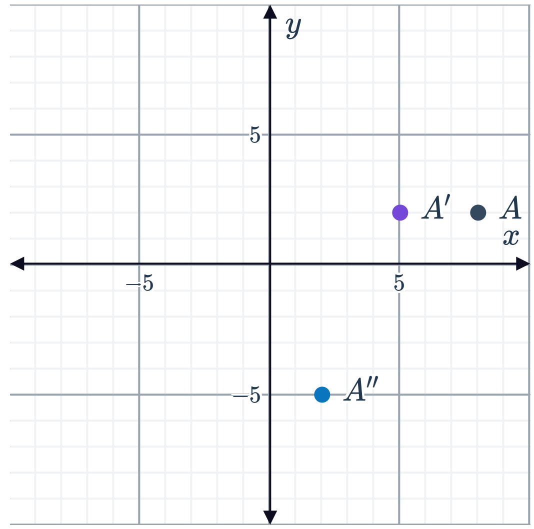
Compositions of transformations

1

a

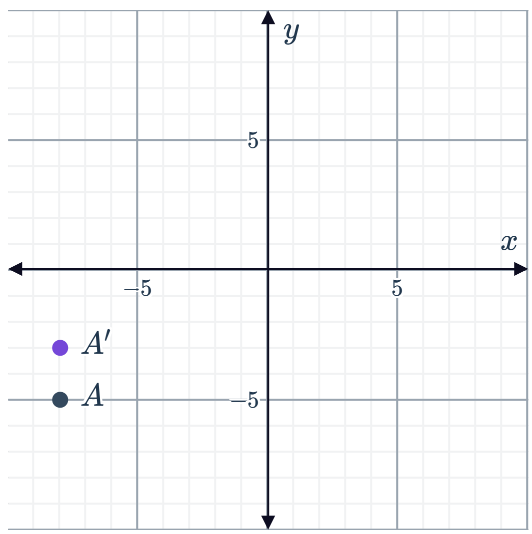


b



2

a



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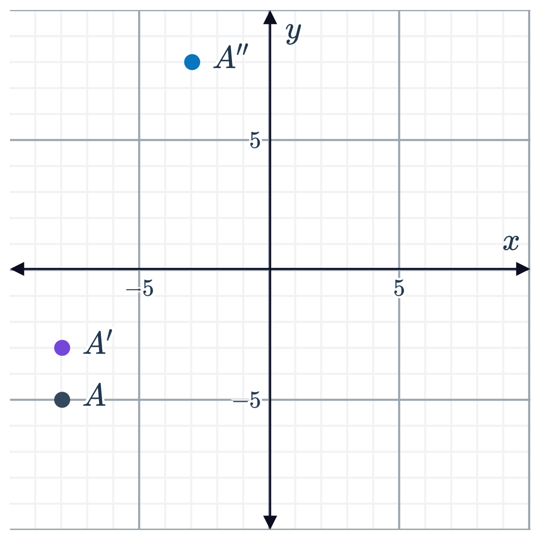
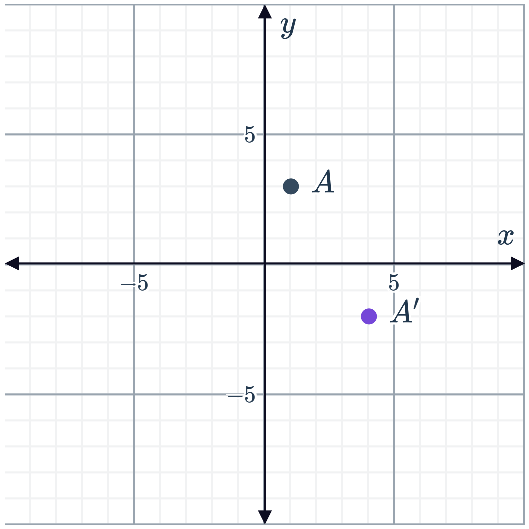


3

a



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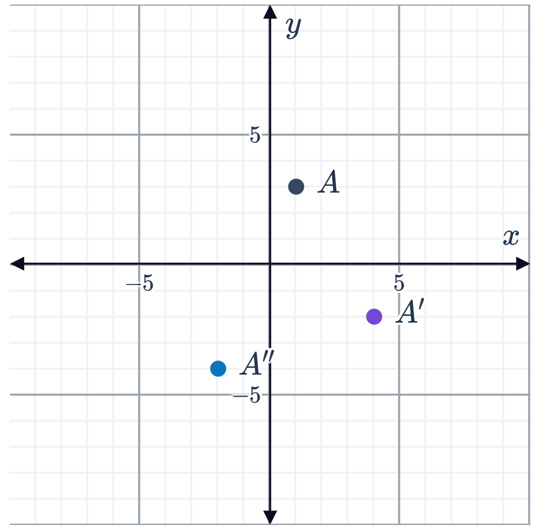

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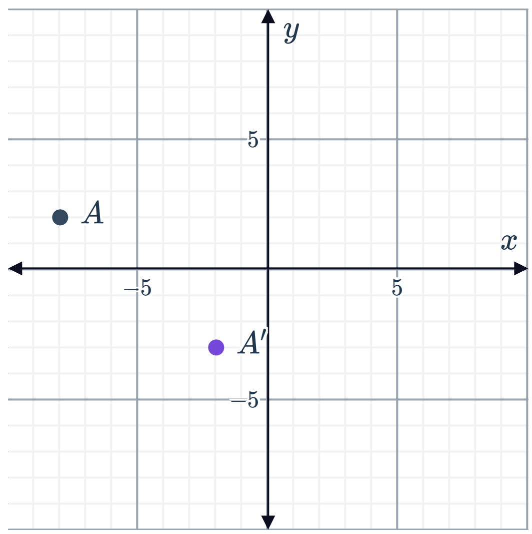

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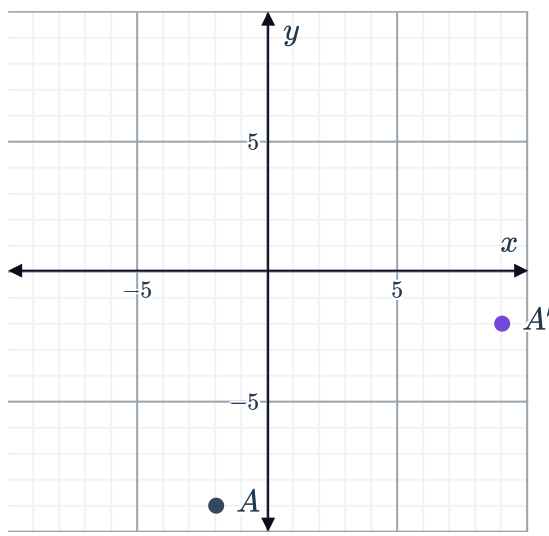

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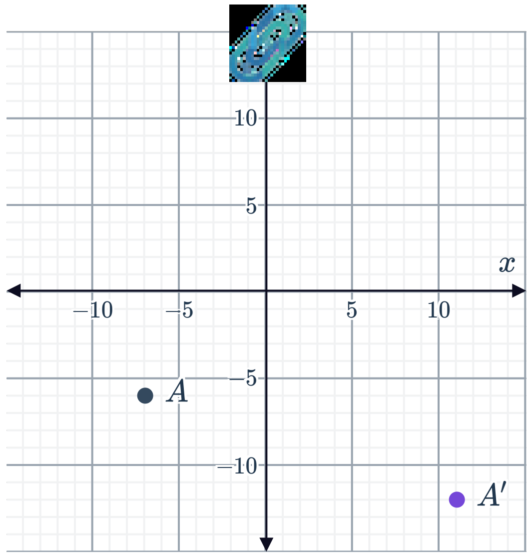


7

a $B(-4,-4)$

b $y = 0$

8

a $(-a,-b)$

b True

9

a $B(7,9)$

b 16 units to the left

10

a $(-3,9)$

b $(-4,9)$

c $(3,-4)$

d $(-10,3)$

11

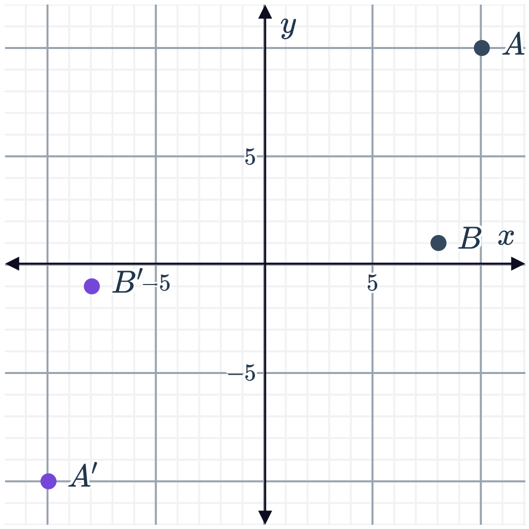
a $A'(-6,-1)$

b

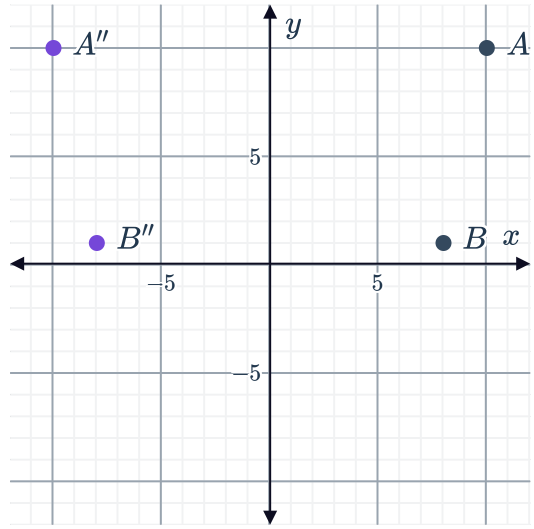
i Yes ii No iii No iv No

12

a



b

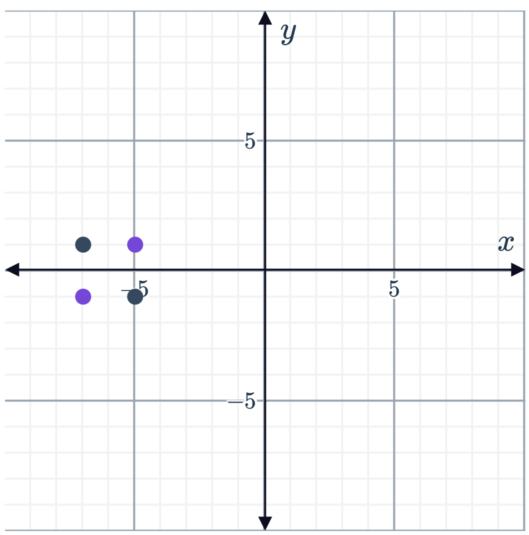


c

i No **ii** Yes **iii** No **iv** No

13

a



b

i No **ii** No **iii** No **iv** Yes

14

a $A'\,(-10, 8)$

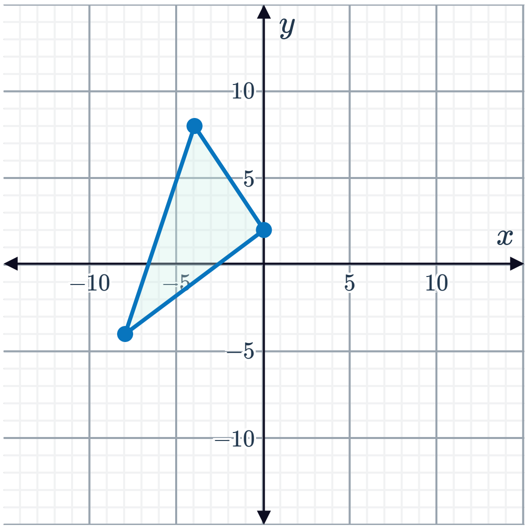
$B'\,(-7, 7)$

b

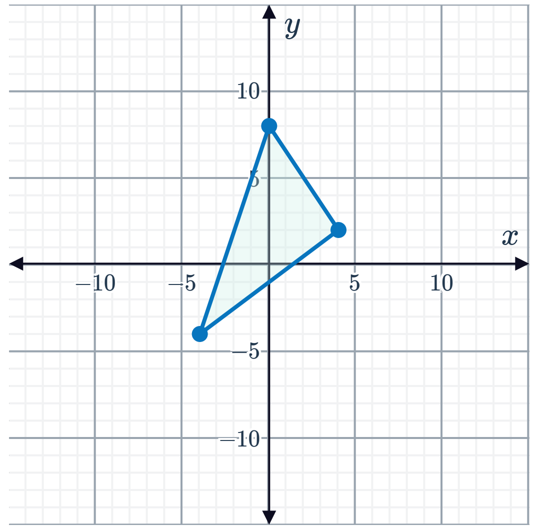
i No **ii** Yes **iii** No **iv** No

15

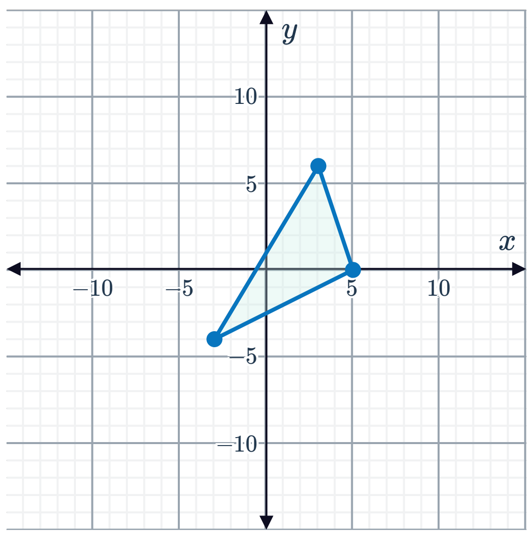
a


y
10
5
x
−10
−5
5
10
−5
−10

b

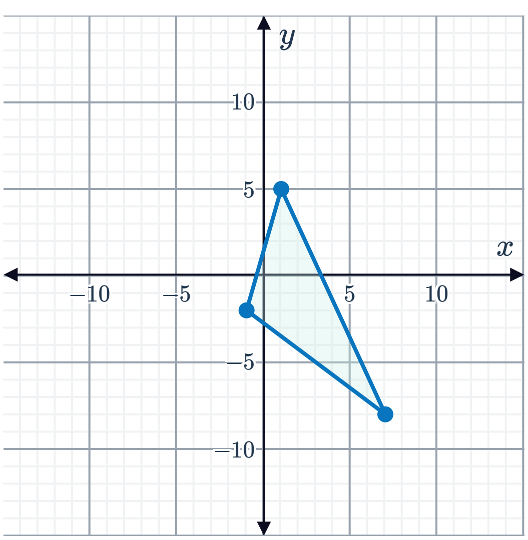

y
10
5
x
−10
−5
5
10
−5
−10

16

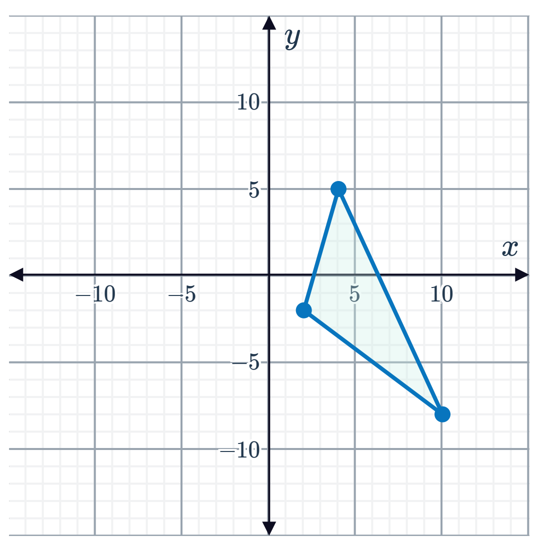

y
10
5
x
−10
−5
5
10
−5
−10

17

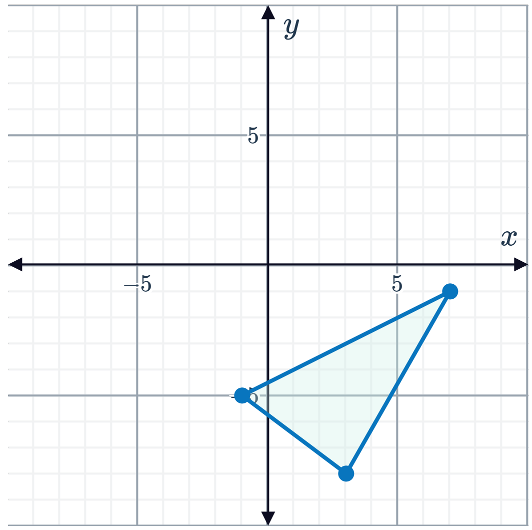
a


y
10
5
x
−10
−5
5
10
−5
−10

b

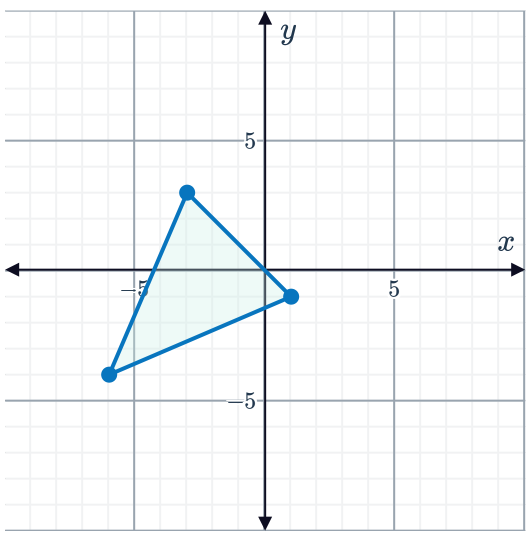

y
10
5
x
−10
−5
5
10
−5
−10

18

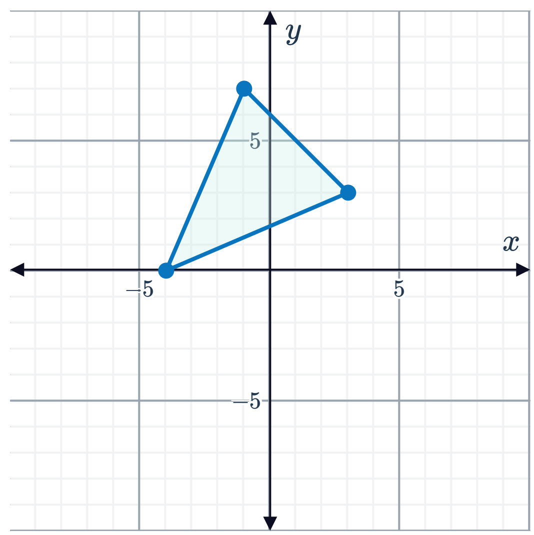


19

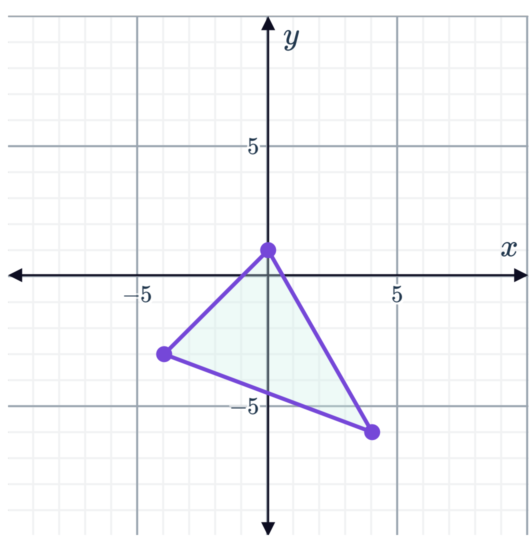
a



b

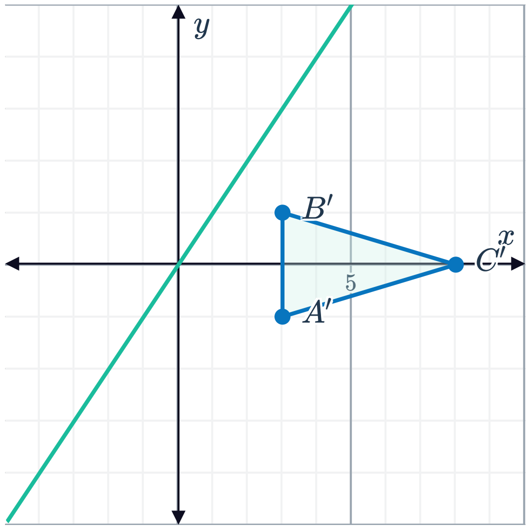


20



21

a



b Possible answers:
A 90° rotation counterclockwise about the origin.
A 270° rotation clockwise about the origin.

22 $A'\ (-3,-5)$
$B'\ (-1,8)$
$C'\ (14,10)$

23 $A'\ (-10,-7)$
$B'\ (5,2)$
$C'\ (5,-1)$

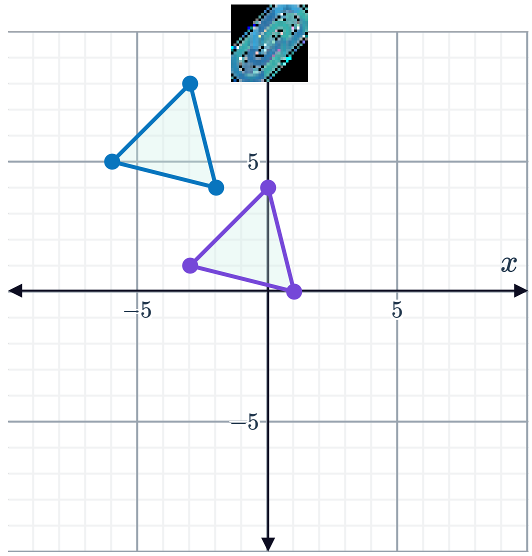
24 $A'\ (1,-9)$
$B'\ (1,3)$
$C'\ (13,2)$

25 $A'\ (-8,-1)$
$B'\ (-5,9)$
$C'\ (-9,12)$

26 $A'\ (4,6)$
$B'\ (-1,5)$
$C'\ (13,-12)$

27 $A'\ (8,-2)$
$B'\ (-1,9)$
$C'\ (-6,6)$

28



29

a No

b No

c No

d Yes

30

a No

b No

c No

d Yes

31

a No

b No

c Yes

d No

32

a

$A(-2, 2)$	$\rightarrow$	$A'(3, -2)$
$B(4, -3)$	$\rightarrow$	$B'(9, -7)$
$C(-4, -3)$	$\rightarrow$	$C'(1, -7)$

b

$A'(3, -2)$	$\rightarrow$	$A''(2, 3)$
$B'(9, -7)$	$\rightarrow$	$B''(7, 9)$
$C'(1, -7)$	$\rightarrow$	$C''(7, 1)$

33

a

$A(-7, -2)$	$\rightarrow$	$A'(-2, 7)$
$B(-3, -2)$	$\rightarrow$	$B'(-2, 3)$
$C(-3, -8)$	$\rightarrow$	$C'(-8, 3)$
$D(-7, -8)$	$\rightarrow$	$D'(-8, 7)$

b

$A'(-2, 7)$	$\rightarrow$	$A''(2, 7)$
$B'(-2, 3)$	$\rightarrow$	$B''(2, 3)$
$C'(-8, 3)$	$\rightarrow$	$C''(8, 3)$
$D'(-8, 7)$	$\rightarrow$	$D''(8, 7)$

34

a No

b Yes

c No

d No

35

a No b No c Yes d No

36

a No b No c No d Yes

37

a No b No c No d Yes

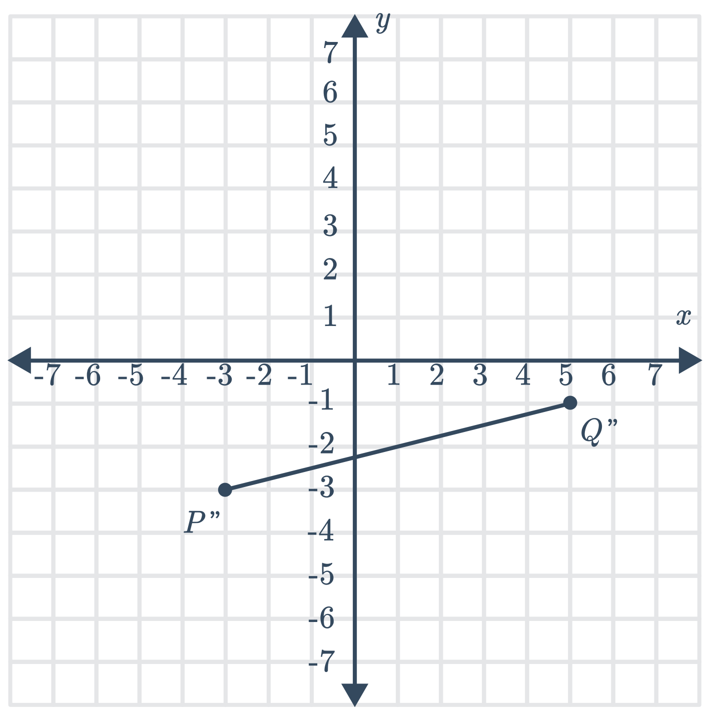
38

a Yes b No c No d No

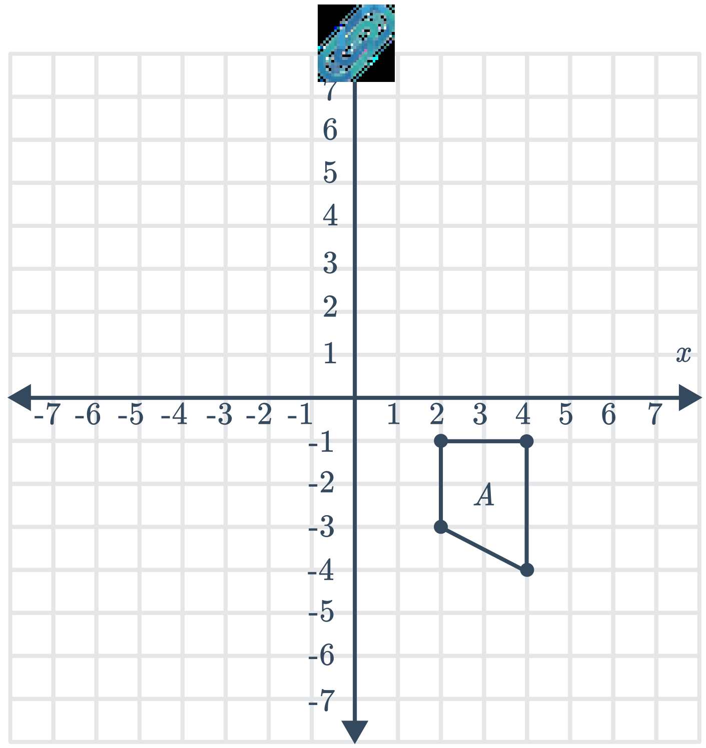
39 $(18,-18),(18,0),(0,0),(0,-18)$

Additional questions

40



41

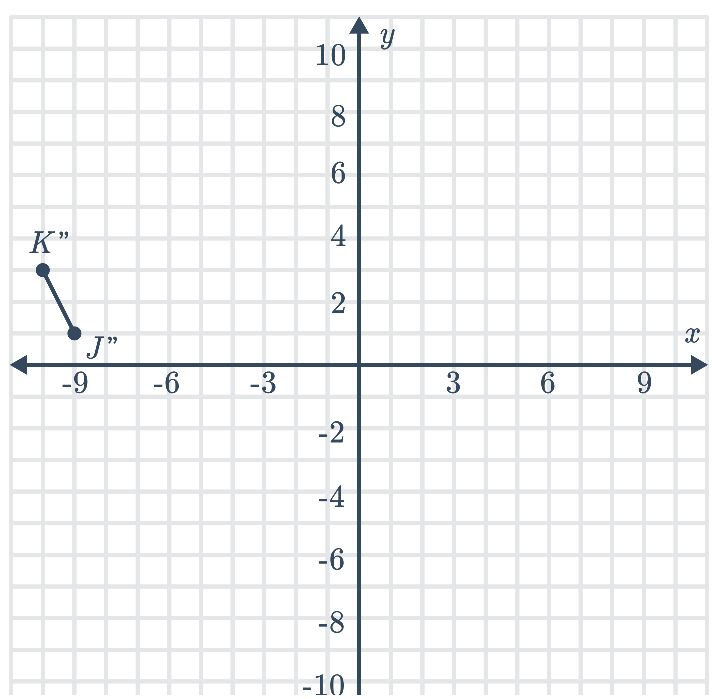


42 Reflect $\triangle ABC$ across the x-axis, then translate $\triangle A'B'C'$ by 2 units up and 6 units to the right.

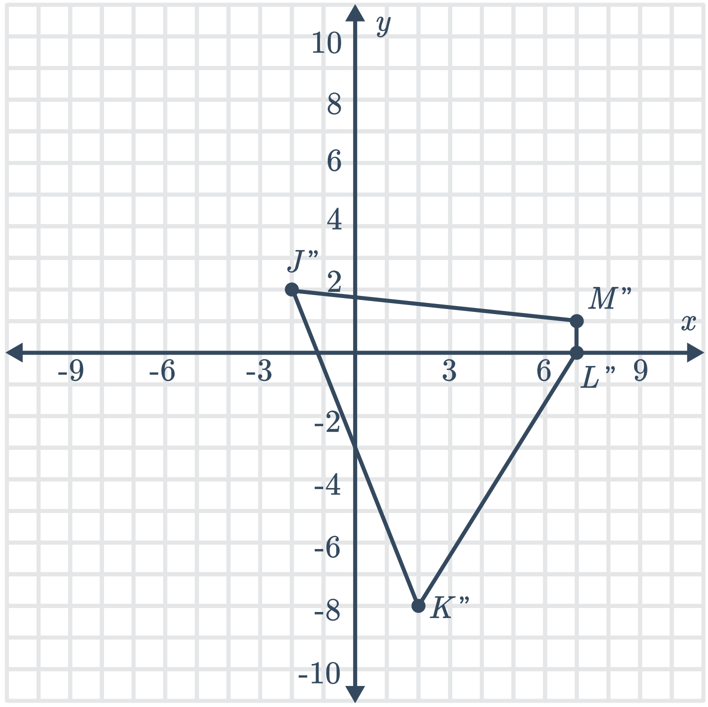
43 Reflect $\triangle ABC$ across the x-axis, then rotate $\triangle A'B'C'$ by 90° counterclockwise about the origin.

44 Translate 1 unit down then reflect $\triangle ABC$ across the line $y = x$.

45



46



47

- **a** Rotate section 1 180° clockwise or counterclockwise about the center of the flag.
- **b** Reflect section 1 over the horizontal midline, then the vertical midline (in either order).

48 Answers may vary. Starting with the top leftmost parallelogram in the pattern, perform a series of translations down the fabric until you have six parallelograms. Then, reflect the set of six parallelograms across the line of symmetry in the pattern.